



قواعد البيانات المتقدمة

12:00-1:00

27/11/2024

د/ابراهيم العوضى

Faculty of Computers & Information, Assiut University

Course code: IS411

4th Level

Mid-term Exam

Duration: one hour

This exam for the following program(s) : (Add the program's name(s))

Points: -/25

1

* الإسم الرباعي (بالعربي فقط) 

2

* رقم الجلوس 

Dec	Hex	Name	Char	Ctrl-char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	0	Null	NUL	CTRL-@	32	20	Space	64	40	@	96	60	`
1	1	Start of heading	SOH	CTRL-A	33	21	!	65	41	A	97	61	a
2	2	Start of text	STX	CTRL-B	34	22	"	66	42	B	98	62	b
3	3	End of text	ETX	CTRL-C	35	23	#	67	43	C	99	63	c
4	4	End of xmit	EOT	CTRL-D	36	24	\$	68	44	D	100	64	d
5	5	Enquiry	ENQ	CTRL-E	37	25	%	69	45	E	101	65	e
6	6	Acknowledge	ACK	CTRL-F	38	26	&	70	46	F	102	66	f
7	7	Bell	BEL	CTRL-G	39	27	'	71	47	G	103	67	g
8	8	Backspace	BS	CTRL-H	40	28	(	72	48	H	104	68	h
9	9	Horizontal tab	HT	CTRL-I	41	29	)	73	49	I	105	69	i
10	0A	Line feed	LF	CTRL-J	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	VT	CTRL-K	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	FF	CTRL-L	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage feed	CR	CTRL-M	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	SO	CTRL-N	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	SI	CTRL-O	47	2F	/	79	4F	O	111	6F	o
16	10	Data line escape	DLE	CTRL-P	48	30	0	80	50	P	112	70	p
17	11	Device control 1	DC1	CTRL-Q	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	DC2	CTRL-R	50	32	2	82	52	R	114	72	r
19	13	Device control 3	DC3	CTRL-S	51	33	3	83	53	S	115	73	s
20	14	Device control 4	DC4	CTRL-T	52	34	4	84	54	T	116	74	t
21	15	Neg acknowledge	NAK	CTRL-U	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	SYN	CTRL-V	54	36	6	86	56	V	118	76	v
23	17	End of xmit block	ETB	CTRL-W	55	37	7	87	57	W	119	77	w
24	18	Cancel	CAN	CTRL-X	56	38	8	88	58	X	120	78	x
25	19	End of medium	EM	CTRL-Y	57	39	9	89	59	Y	121	79	y
26	1A	Substitute	SUB	CTRL-Z	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	ESC	CTRL-[	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	FS	CTRL-\	60	3C	<	92	5C	\	124	7C	
29	1D	Group separator	GS	CTRL-]	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	RS	CTRL-^	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	US	CTRL-`	63	3F	?	95	5F	_	127	7F	DEL

8

ASCII code ☐

9

What is the key value for the text **WE LOVE EGYPT** 

	Col A	Col B	Col C
Row #0	a0	b0	c0
Row #1	a1	b1	c1
Row #2	a2	b2	c2
Row #3	a3	b3	c3
Row #4	a4	b4	c4
Row #5	a5	b5	c5

10

The storage model of the next figure is

NSM

DSM

PAX

Page #1	header	null bitmap				
	a0	a1	a2	a3	a4	a5
Page #2	header	null bitmap				
	b0	b1	b2	b3	b4	b5
Page #3	header	null bitmap				
	c0	c1	c2	c3	c4	
	c5					

11

This figure refers to 

Strawman Idea

Slotted pages

Tuble layout

Page
Num Tuples = 3
Tuple #1
Tuple #2
Tuple #3

Record ID

12

Page size in MySQL is 

8 KB

4 KB

1 KB

16 KB

13

Denormalized tube data is 

describes how tables are stored in pages.

describes how related tables are stored in pages.

avoided in database, and should normalized by normalization forms

is important topic in ER-model

14

The data structures used in database are RAM data structure 

True

False

15

Database is considered as sert of files 

True

False

16

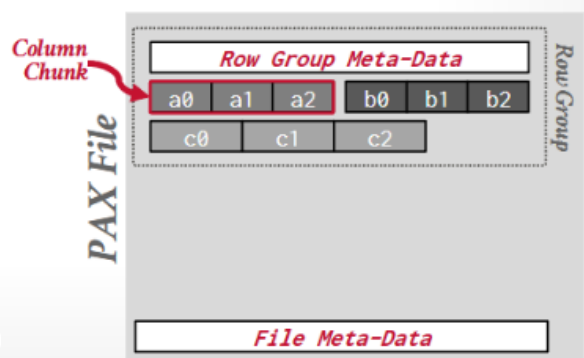
Assume that you have 10,000,000,000 values, and B+ tree node has 5 pointers.
What is the highest depth of this + tree? 

	Col A	Col B	Col C
Row #0	a0	b0	c0
Row #1	a1	b1	c1
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17

The storage model of the next figure is

NSM



DSM

PAX

18

To print an interger variable, which hase the name number, in a text file, we use

..... 

```
fprintf(fPtr, "%d", number);
```

```
fwrite(&number, sizeof(int), 1, fPtr);
```

```
fwrite(fPtr, "%d", number);
```

19

Assume that you have a hash table with ten element. Where the text **WE LOVE EGYPT** can be stored in this hash table? 

20

This code is from a very important book. **Even this code is correct, It is is not recommended to write as this code.** 

True

False

```
14) $q = $d->query('INSERT INTO EMPLOYEE VALUES (?, ?, ?, ?)',
15) array($eid, $_POST['emp_name'], $_POST['emp_job'], $_POST['emp_dno']));
```

21

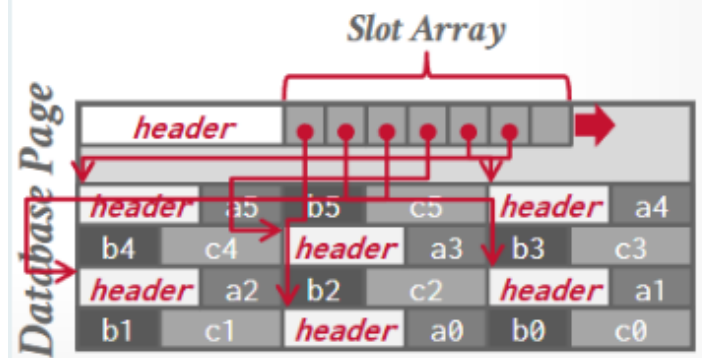
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Row #3	a3	b3	c3
Row #4	a4	b4	c4
Row #5	a5	b5	c5



22

PHP is language

low-level

high language

scripting language

23

Every page contains a header of meta-data about the page's contents.

Page Size

Checksum

DBMS Version

Transaction Visibility

Compression / Encoding Meta-data

Schema Information

Data Summary / Sketches

For any page storage architecture, we now need to decide how to organize the data inside of the page.

24

PHP file should be into 

one part

two parts

three parts



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923 ✓

859 ✓

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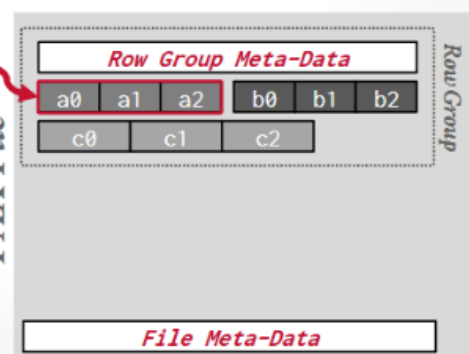
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Column
Chunk

PAX File

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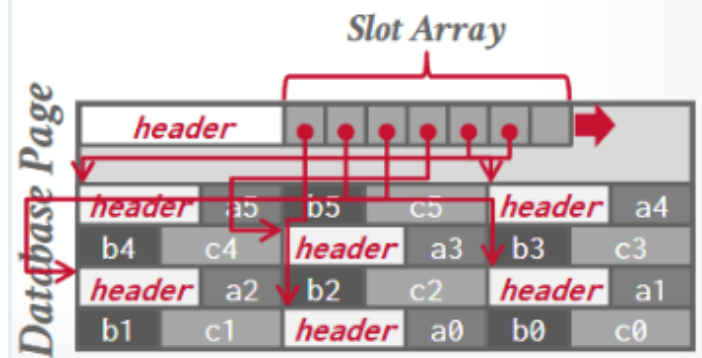
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